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United States Patent	12397447
Kind Code	B2
Date of Patent	August 26, 2025
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Grip device and robot device comprising same

Abstract

A grip device is provided. A grip device according to an embodiment of the present disclosure includes: a first finger; a second finger facing the first finger; a first link part including a first guide slot and supporting the first finger; a second link part supporting the second finger and including a second guide slot, intersecting the first link part; a hinge configured to move inside the first guide slot and second guide slot and connecting the first link part and the second link part at an intersection point of the first link part and second link part; a first actuator configured to adjust a distance between the first finger and second finger by moving the first link part and/or the second link part; and a second actuator configured to move the hinge inside the first guide slot and second guide slot.

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Appl. No.: 18/074018

Filed: December 02, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230102104 A1	Mar. 30, 2023

Foreign Application Priority Data

KR	10-2020-0092103	Jul. 24, 2020
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Related U.S. Application Data

continuation parent-doc WO PCT/KR2021/008350 20210701 PENDING child-doc US 18074018

Publication Classification

Int. Cl.: B25J15/02 (20060101); B25J9/16 (20060101); B25J13/08 (20060101)

U.S. Cl.:

CPC B25J15/0266 (20130101); B25J9/1612 (20130101); B25J13/08 (20130101);

Field of Classification Search

CPC: B25J (15/0266); B25J (15/028); B25J (9/1612); B25J (9/1694); B25J (9/102); B25J (9/126); B25J (13/08); B25J (13/082); B25J (13/086); B25J (19/023); G05B (2219/39543)

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS (1) This application is a bypass continuation of International Application No. PCT/KR2021/008350, filed on Jul. 1, 2021, in the

Korean Patent Office, which claims priority from Korean Application No. 10-2020-0092103, filed on Jul. 24, 2020, in the Korean Patent Office, the contents of which are herein incorporated by reference in their entireties.

BACKGROUND

1. Technical Field

(1) Embodiments of the disclosure relate to a grip device and a robot device including the same, and more particularly, to a grip device having an improved structure in which a hinge functioning as a support point is moved to obtain a force transmission ratio suitable to grip an object, and a robot device including the same.

2. Background Art

(2) With the development of electronic technology, various types of electronic devices have been developed. In particular, various types of robot devices for performing tasks in place of humans at industrial sites or in the fields of medical treatment, space, housework, etc. have recently been developed. Such a robot device may include a grip device to perform various tasks such as gripping an object, assembly, transfer, welding, etc.

(3) It is important to obtain a large grip force while reducing a gap between the fingers of the grip device in order for the grip device to grip a flat object such as a dish. On the contrary, it is important to obtain a small grip force while increasing a gap between the fingers of the grip device in order for the grip device to grip a tall object such as a cup.

(4) However, a grip device of the related art performs a grip operation uniformly regardless of the shape of an object to be gripped and thus is driven in an inefficient manner. In addition, a driving speed of a grip device using a gear method or a screw rotation method is low.

SUMMARY

(5) To address the above-described problems, embodiments of the disclosure provide a grip device having an improved structure in which a hinge functioning as a support point is moved to obtain a force transmission rate suitable to grip an object, and a robot device including the same.

(6) According to embodiments of the disclosure, a grip device is provided. The grip device may include: a first finger; a second finger facing the first finger; a first link part including a first guide slot and configured to support the first finger; a second link part including a second guide slot, the second link part crossing the first link part and configured to support the second finger; a hinge configured to be moved within the first guide slot and the second guide slot and connect the first link part and the second link part at an intersection of the first link part and the second link part; a first actuator configured to move at least one of the first and the second link part to adjust a gap between the first finger and the second finger; and a second actuator configured to move the hinge within the first guide slot and the second guide slot.

(7) According to an embodiment of the disclosure, the first link part may further include: a first linkage connected to a first point on the first finger; a second linkage connected to a second point on the first finger; and a third linkage connected to the second linkage and provided in parallel with the first finger, the second link part may further include: a fourth linkage connected to a first point on the second finger; a fifth linkage connected to a second point on the second finger and provided in parallel with the fourth linkage; and a sixth linkage connected to the fifth linkage and provided in parallel with the second finger, and the first guide slot is formed in the second linkage, and the second guide slot is formed in the fifth linkage.

(8) According to an embodiment of the disclosure, a length of the third linkage may be equal to a distance between the first point and the second point on the first finger, and a length of the sixth linkage may be equal to a distance between the first point and the second point on the second finger.

(9) According to an embodiment of the disclosure, the first actuator may be configured to adjust a gap between the third linkage and the sixth linkage.

- (10) According to an embodiment of the disclosure, the first actuator may include: a first driving motor configured to move the third linkage in a first direction; and a second driving motor configured to move the sixth linkage in a second direction opposite to the first direction.
- (11) According to an embodiment of the disclosure, the second actuator may include a driving motor configured to move the hinge within the first guide slot and the second guide slot.
- (12) According to an embodiment of the disclosure, the second actuator may include: a first driving motor provided on the second linkage and configured to move the hinge along the first guide slot; and a second driving motor provided on the fifth linkage and configured to move the hinge along the second guide slot.
- (13) According to an embodiment of the disclosure, each of the first driving motor and the second driving motor may include a Scotch yoke type driving motor, a rack and pinion type driving motor, or a hydraulic type driving motor.
- (14) According to an embodiment of the disclosure, the second linkage and the fifth linkage may each have a convex shape with respect to a space between the first finger and the second finger.
- (15) According to embodiments of the disclosure, a robot device is provided. The robot device may include: an image sensor configured to detect at least one of a shape and position of an object; a grip device including a link part, the grip device configured to adjust a support point of the link part by moving a hinge on the link part, wherein the link part is cross-arranged with respect to a first finger of the grip device, a second finger of the grip device, and the hinge, and the link part is configured to support the first finger and the second finger; and a processor configured to control the grip device to position the hinge to correspond to the object detected by the image sensor.
- (16) According to an embodiment of the disclosure, the link part may include: a first link part including a first guide slot and configured to support the first finger; and a second link part including a second guide slot, the second link part crossing the first link part and configured to support the second finger, the hinge may be movable within the first guide slot and the second guide slot and may connect the first link part and the second link part at an intersection of the first link part and the second link part, and the grip device may further include an actuator configured to move the hinge within the first guide slot and the second guide slot.
- (17) According to an embodiment of the disclosure, the actuator may include a connection member which is connected to the hinge and is variable in length, and the processor may be configured to determine a target length of the connection member according to at least one of the shape and position of the object detected by the image sensor, and control the actuator to cause a length of the connection member to be equal to the target length that is determined.
- (18) According to an embodiment of the disclosure, the robot device may further include a pressure sensor configured to measure pressure applied to at least one of the first finger and the second finger, and wherein the processor may be configured to control a grip operation of the grip device based on a value measured by the pressure sensor.
- (19) According to an embodiment of the disclosure, the robot device may further include: a first distance sensor on the first finger; and a second distance sensor on the second finger, and wherein the processor may be further configured to: activate the second distance sensor based on the first distance sensor detecting the object during movement of the robot device; receive information about distances to the object detected by the first distance sensor and the second distance sensor at a plurality of points according to the movement of the robot device; and control a grip operation of the grip device based on the information that is received.
- (20) According to embodiments of the disclosure, a method of operating a robot device is provided. The method may include: detecting, by an image sensor of the robot device, at least one of a shape and position of an object; and controlling, by at least one processor of the robot device, a grip device of the robot device to position a hinge on a link part of the grip device to correspond to the object detected by the image sensor, wherein the grip device may be configured to adjust a support point of the link part by moving the hinge on the link part, wherein the link part may be cross-

arranged with respect to a first finger of the grip device, a second finger of the grip device, and the hinge, and the link part may be configured to support the first finger and the second finger.

(21) According to an embodiment of the disclosure, the link part may include: a first link part including a first guide slot and configured to support the first finger; and a second link part including a second guide slot, the second link part crossing the first link part and configured to support the second finger, wherein the controlling the grip device may include controlling an actuator of the robot device to move the hinge within the first guide slot and the second guide slot, and wherein the hinge may connect the first link part and the second link part at an intersection of the first link part and the second link part.

(22) According to an embodiment of the disclosure, the actuator may include a connection member which is connected to the hinge and is variable in length, and the controlling the grip device may further include determining a target length of the connection member according to at least one of the shape and position of the object detected by the image sensor, and controlling the actuator to cause a length of the connection member to be equal to the target length that is determined.

(23) According to an embodiment of the disclosure, the controlling the grip device may include controlling, by the at least one processor, a grip operation of the grip device based on a value measured by a pressure sensor of the robot device, and the pressure sensor may be configured to measure pressure applied to at least one of the first finger and the second finger.

(24) According to an embodiment of the disclosure, the method may further include: activating, by the at least one processor, a second distance sensor, on the second finger, based on a first distance sensor, on the first finger, detecting the object during movement of the robot device; and receiving information about distances to the object detected by the first distance sensor and the second distance sensor at a plurality of points according to the movement of the robot device, wherein controlling the grip device may include controlling a grip operation of the grip device based on the information that is received.

(25) According to an embodiment of the disclosure, the controlling the grip operation may include controlling the grip operation based on a comparison of a difference, between a distance between the first distance sensor and the object and a distance between the second distance sensor and the object, and a threshold value.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a side view of a grip device according to an embodiment of the disclosure.

(2) FIG. 2 is a side view of the grip device of FIG. 1 in which a hinge is moved backward.

(3) FIG. 3 is a side view of a grip device in which lengths of a third driving motor and a fourth driving motor are different from each other.

(4) FIG. 4 is a side view of a grip device in which lengths of a third driving motor and a fourth driving motor are different from each other.

(5) FIG. 5 is a side view of a grip device in which each of first and second link parts is configured as a single link.

(6) FIG. 6 is a block diagram of a robot device according to an embodiment of the disclosure.

(7) FIG. 7A is a first diagram for describing a process in which a grip operation of a grip device is controlled according to a value measured by a distance sensor.

(8) FIG. 7B is a second diagram for describing the process in which the grip operation of the grip device is controlled according to the value measured by the distance sensor.

(9) FIG. 7C is a third diagram for describing the process in which the grip operation of the grip device is controlled according to the value measured by the distance sensor.

(10) FIG. 7D is a fourth diagram for describing the process in which the grip operation of the grip

device is controlled according to the value measured by the distance sensor.

(11) FIG. 7E is a fifth diagram for describing the process in which the grip operation of the grip device is controlled according to the value measured by the distance sensor.

(12) FIG. 7F is a sixth diagram for describing the process in which the grip operation of the grip device is controlled according to the value measured by the distance sensor.

DETAILED DESCRIPTION

(13) Embodiments described below are provided as non-limiting examples to help understand the disclosure, and it should be understood that embodiments of the disclosure may be implemented in various forms different from these embodiments. In the following description of the disclosure, related well-known functions or components are not described in detail and are not illustrated in the drawings when it is determined that they would obscure the subject matter of the disclosure due to unnecessary detail. In the accompanying drawings, components are not shown in actual scale and the sizes of some components may be exaggerated to help understand the disclosure.

(14) In the present specification and the claims, general terms are selected in consideration of the functions of the disclosure. However, non-general terms may be selected according to the intention of the technician in the art, legal or technical interpretation, the emergence of new technologies, etc. Some terms may be arbitrarily selected by the applicant. These terms may be interpreted as defined in the present specification, and may be interpreted based on the overall content of the present specification and the common technical knowledge in the technical field concerned when the terms are not specifically defined herein.

(15) As used herein, expressions such as “have”, “may have,” “include” or “may include” are intended to indicate the presence of features (e. g., a numerical value, a function, an operation, a component of a machine part, etc.) and do not exclude the presence of additional features.

(16) While the present specification describes components for describing example embodiments of the disclosure, the disclosure is not necessarily limited thereto. Accordingly, some components may be changed or omitted and other components may be added. In addition, components may be distributed and disposed in different independent devices.

(17) Although example embodiments of the disclosure will be described in detail herein with reference to the accompanying drawings and the content shown in the accompanying drawings, the disclosure is not limited to or limited by the example embodiments.

(18) Hereinafter, embodiments of the disclosure will be described in more detail with reference to the accompanying drawings.

(19) FIG. 1 is a side view of a grip device according to an embodiment of the disclosure.

(20) Referring to FIG. 1, a grip device **1** according to an embodiment of the disclosure may include a first finger **10**, a second finger **20**, a first link part **100**, a second link part **200**, a hinge **300**, a first actuator **400**, a second actuator **500**, and a support **600**.

(21) The first finger **10** and the second finger **20** are disposed to face each other and are spaced a certain distance from each other to form a space for disposing therein an object to be gripped.

(22) The first finger **10** and the second finger **20** may be driven by the first actuator **400** and the second actuator **500**, which will be described below, and a gap L2 between the first finger **10** and the second finger **20** may be adjusted by a driving force transmitted by the first link part **100** and the second link part **200**.

(23) The first link part **100** may be provided with a first guide slot **121** and may support the first finger **10**. The second link part **200** may be provided with a second guide slot **221** and may support the second finger **20**.

(24) Each of the first link part **100** and the second link part **200** may have a four-bar linkage structure. Specifically, each of the first link part **100** and the second link part **200** may include three linkages connected to one another in sequence, and a total of four contact points among the three linkages and the first finger **10** or the second finger **20** are connected to form a closed-loop shape.

(25) However, the first link part **100** and the second link part **200** are not limited to the above-

described structure and may be each embodied as single link. For example, a single linkage of the first link part **100** and a single linkage of the second link part **200** may support the first finger **10** and the second finger **20** while being cross-connected with each other.

(26) The hinge **300** is movable within the first guide slot **121** and the second guide slot **221** and may connect the first link part **100** and the second link part **200** at an intersection of the first link part **100** and the second link part **200**.

(27) The hinge **300** may function as a support point of the first link part **100** and the second link part **200**. That is, the first link part **100** and the second link part **200** may rotate around the hinge **300** as a support point, thereby performing a grip operation.

(28) Because the hinge **300** is movable within the first guide slot **121** and the second guide slot **221**, a position of the hinge **300** may be changed to be close to or away from an object to be gripped.

(29) Thus, in order to grip an object to be gripped with a large grip force, the hinge **300** may be disposed close to the object to increase a force transmission ratio and thus the object may be effectively gripped with a small driving force. In contrast, in order to grip an object to be gripped with a small grip force, the hinge **300** may be disposed away from the object to reduce a force transmission ratio and thus the object may be easily gripped with the same driving force without changing the intensity of the driving force.

(30) The first actuator **400** may move at least one of the first link part **100** and the second link part **200** to adjust a gap L2 between the first finger **10** and the second finger **20**. The first actuator **400** may adjust a gap L1 between a third linkage **130** and a sixth linkage **230** to be described below.

(31) Specifically, because the first actuator **400** increases or decreases the gap L1 between one end of the first link part **100** and one end of the second link part **200** and the first link part **100** and the second link part **200** rotate around the hinge **300**, the gap L2 between the first finger **10** and the second finger **20** respectively supported by the first link part **100** and the second link part **200** may also increase or decrease.

(32) When the grip device **1** performs the grip operation, the first actuator **400** may move one end of the first link part **100** upward and one end of the second link part **200** downward. Thus, the first link part **100** may rotate counterclockwise around the hinge **300**, and the second link part **200** may rotate clockwise around the hinge **300**. As the first link part **100** and the second link part **200** rotate, the first finger **10** and the second finger **20** may be moved close to each other to pressurize and grip an object therebetween.

(33) Conversely, when the grip device **1** performs a drop operation, the first actuator **400** may move one end of the first link part **100** downward and one end of the second link part **200** upward. Thus, the first link part **100** may rotate clockwise around the hinge **300**, and the second link part **200** may rotate counterclockwise around the hinge **300**. As the first link part **100** and the second link part **200** rotate, the first finger **10** and the second finger **20** may be moved away from each other to drop an object therebetween.

(34) The first actuator **400** may include a first driving motor **410** configured to move the third linkage **130** of the first link part **100** in a first direction and a second driving motor **420** configured to move the sixth linkage **230** of the second link part **200** in a second direction opposite to the first direction. The first driving motor **410** and the second driving motor **420** may be linear motors but are not limited thereto and the type of the first driving motor **410** and the second driving motor **420** is not limited as long as they are capable of moving the third linkage **130** and the sixth linkage **230**.

(35) That is, as the first driving motor **410** (e.g., a linear motor) and the second driving motor **420** (e.g., a linear motor) move the third linkage **130** and the sixth linkage **230** in opposite directions, the first finger **10** and the second finger **20** may perform the grip operation or the drop operation.

(36) Although the first actuator **400** is illustrated as including the first driving motor **410** to move the first link part **100** and the second driving motor **420** to move the second link part **200**, the first actuator **400** is not limited thereto. For example, the first actuator **400** may include only the first linear motor **410** and one end of the second link part **200** may be fixed on the support **600**, so that

the grip operation and the drop operation may be smoothly performed by moving only the first link part **100**.

(37) The second actuator **500** may move the hinge **300** within the first guide slot **121** and the second guide slot **221**. Specifically, the second actuator **500** may include only a third driving motor **510** to move the hinge **300** or may include both the third driving motor **510** and a fourth driving motor **520** to move the hinge **300**.

(38) The third driving motor **510** may include a main body **511** and a connection member **512**. One end of the connection member **512** of the third driving motor **510** may be connected to the hinge **300** and a length of the connection member **512** may be variable.

(39) Specifically, the length of the connection member **512** may be changed to be away from or close to the main body **511** and thus the hinge **300** connected to the connection member **512** may be moved along the first guide slot **121** and the second guide slot **221**.

(40) Although the third driving motor **510** is illustrated as being disposed on a second linkage **120** of the first link part **100**, embodiments are not limited thereto and the main body **511** may be fixed on the support **600** at the same height as the hinge **300** to move the hinge **300** forward or backward.

(41) The fourth driving motor **520** may have the same structure as the third driving motor **510** described above. That is, the fourth driving motor **520** may include a main body **521** and a connection member **522**, one end of the connection member **522** may be connected to the hinge **300**, and a length of the connection member **522** may be variable to be away from or close to the main body **521**.

(42) The third driving motor **510** may be disposed on the second linkage **120** of the first link part **100** to move the hinge **300** along the first guide slot **121**. The fourth driving motor **520** may be disposed on a fifth linkage **220** of the second link part **200** to move the hinge **300** along the second guide slot **221**.

(43) The connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** may be moved to have the same length or different lengths. Accordingly, the hinge **300** may be moved not only forward or backward but also upward or downward, and an area to be gripped by the grip device **1** may be expanded. This will be described in detail with reference to FIGS. **3** and **4** below.

(44) The third driving motor **510** and the fourth driving motor **520** may be linear motors each having a variable length, and may be embodied as a scotch yoke type, a rack and pinion type, or a hydraulic type but are not limited thereto.

(45) Hereinafter, the structures of the first link part **100** and the second link part **200** will be described in more detail.

(46) The first link part **100** may include a first linkage **110** connected to a first point **11** on the first finger **10**, a second linkage **120** connected to a second point **12** on the first finger **10** and disposed in parallel with the first linkage **110**, and a third linkage **130** connected to the second linkage **120** and disposed in parallel with the first finger **10**.

(47) The second link part **200** may include a fourth linkage **210** connected to a first point **21** on the second finger **20**, a fifth linkage **220** connected to a second point **22** on the second finger **20** and disposed in parallel with the fourth linkage **210**, and a sixth linkage **230** connected to the fifth linkage **220** and disposed in parallel with the second finger **20**.

(48) The third linkage **130** and the sixth linkage **230** may maintain a horizontal shape even when the first link part **100** and the second link part **200** rotate. The first guide slot **121** may be formed in the second linkage **120**, and the second guide slot **221** may be formed in the fifth linkage **220**.

(49) The third linkage **130** may have a length $S1$ equal to a distance $D1$ between the first point **11** and the second point **12** on the first finger **10**. The sixth linkage **230** may have a length $S2$ equal to a distance between the first point **21** and the second point **22** on the second finger **20**.

(50) As the first linkage **110**, the second linkage **120**, and the third linkage **130** have the above-described structure, a region of each of the first linkage **110**, the second linkage **120**, and the third

linkage **130** and the first finger **10** has a shape substantially the same as a parallelogram and thus the first finger **10** may maintain a horizontal shape.

(51) Likewise, as the fourth linkage **210**, the fifth linkage **220**, and the sixth linkage **230** have the above-described structure, a region of each of the fourth linkage **210**, the fifth linkage **220**, and the sixth linkage **230** and the second finger **20** has a shape substantially the same as a parallelogram and thus the second finger **20** may maintain a horizontal shape.

(52) That is, the grip device **1** according to an embodiment of the disclosure has a structure in which a four-bar linkage having a shape of a parallelogram supports the first finger **10** and the second finger **20** symmetrically and thus the first finger **10** and the second finger **20** are capable of gripping objects stably while maintaining a horizontal state.

(53) The second linkage **120** and the fifth linkage **220** may have a convex shape with respect to a space between the first finger **10** and the second finger **20**. That is, a region of each of the second linkage **120** and the fifth linkage **220** may have a convex shape when viewed backward from a space between the first finger **10** and the second finger **20** in which an object to be gripped is placed.

(54) Accordingly, the space between the first finger **10** and the second finger **20** increase and thus even a bulky object may be easily gripped. When the grip operation is performed, an object may be easily gripped while minimizing interference between the object to be gripped and the first link part **100** and the second link part **200**.

(55) The support **600** may be an arm or a body part of a robot, and may support the first actuator **400** and provide a path in which the first actuator **400** moves. Although the support **600** is illustrated as having a vertical shape, the support **600** is not limited thereto.

(56) FIG. **1** illustrates an example in which in the grip device **1**, the connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** are moved to increase a length thereof, so that the hinge **300** may be disposed close to an object to be gripped.

(57) In order to grip a flat object (e. g., a dish) by the grip device **1**, a narrow gap L2 between the first finger **10** and the second finger **20** and a large grip force may be used to grip the flat object. Therefore, as shown in FIG. **1**, by moving the hinge **300** close to the flat object by the third driving motor **510** and the fourth driving motor **520**, the flat object may be easily gripped with a high force transmission ratio while reducing the gap L2 between the first finger **10** and the second finger **20**.

(58) FIG. **2** is a side view of a state in which the hinge **300** of FIG. **1** is moved backward.

(59) Referring to FIG. **2**, as the connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** are moved to reduce the length thereof, compared to FIG. **1**, the hinge **300** may also be moved away from an object to be gripped.

(60) In order to grip a tall object (e. g., a cup) by the grip device **1**, a large gap L2 between the first finger **10** and the second finger **20** and a low grip force may be used to grip the object. Therefore, as shown in FIG. **2**, by moving the hinge **300** away from the tall object by the third driving motor **510** and the fourth driving motor **520**, the object may also be easily gripped with a low force transmission ratio while increasing the gap L2 between the first finger **10** and the second finger **20**.

(61) That is, the grip device **1** according to an embodiment of the disclosure is capable of gripping various types of objects by using the most effective structure by changing the gap L2 between the first finger **10** and the second finger **20** and the force transmission ratio by moving the hinge **300**, which functions as a support point, close to or away from the object, even when the same driving source is used.

(62) FIGS. **3** and **4** are side views of a state in which lengths of the third driving motor **510** and the fourth driving motor **520** are different from each other.

(63) FIG. **3** illustrates an example in which the connection member **512** of the third driving motor **510** is longer than the connection member **522** of the fourth driving motor **520** and thus the hinge **300** is moved upward. In contrast, FIG. **4** illustrates an example in which the connection member

512 of the third driving motor **510** is shorter than the connection member **522** of the fourth driving motor **520** and thus the hinge **300** is moved downward.

(64) Specifically, as the lengths of the third driving motor **510** and the fourth driving motor **520** are different from each other, a grip line G representing a height at which gripping is performed by the first finger **10** and the second finger **20** may be located above or below a center line C of the grip device **1**.

(65) The grip line G is a horizontal line corresponding to a height of the hinge **300** and may represent a point at which an object is to be gripped by the grip device **1**. When the grip device **1** has a vertically symmetrical shape, the center line C may be a horizontal line that is a reference for the symmetry.

(66) For example, when an object to be gripped is placed on a high place such as a shelf, the connection member **512** of the third driving motor **510** may operate to be longer than the connection member **522** of the fourth driving motor **520**. Thus, the grip line G of the grip device **1** is higher than the center line C and thus the first finger **10** and the second finger **20** may easily grip the object on the high place without the shelf interfering.

(67) In contrast, when an object to be gripped is placed on a low place such as a floor or a desk, the connection member **512** of the third driving motor **510** may operate to be shorter than the connection member **522** of the fourth driving motor **520**. Thus, the grip line G of the grip device **1** is lower than the center line C and thus the first finger **10** and the second finger **20** may easily grip the object on the low place without, for example, a floor or a desk interfering.

(68) That is, the grip device **1** according to an embodiment of the disclosure may move such that the lengths of the connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** are equal to each other, and thus, the hinge **300** may be moved to a position corresponding to a shape of an object to be gripped and may easily grip the object even with a low force.

(69) In addition, the grip device **1** according to an embodiment of the disclosure may move such that the lengths of the connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** are different from each other, and thus, the hinge **300** may form a grip line G different from the center line C of the grip device **1** and may easily grip an object without a support, such as a shelf, a floor, or a desk, interfering.

(70) FIG. 5 is a side view of a grip device in which each of first and second link parts is configured as a single link.

(71) Referring to FIG. 5, in a grip device **1a**, a first link part **100** and a second link part **200** configured to respectively support a first finger **10** and a second finger **20** may be each configured as a single link. The first link part **100** and the second link part **200** may be disposed to cross each other and connected to each other through a hinge **300**. For example, the first link part **100** and the second link part **200** may have a roughly shape of scissors.

(72) A first guide slot **101** and a second guide slot **201** may be respectively formed in the first link part **100** and the second link part **200** in a longitudinal direction. The hinge **300** may be moved within the first guide slot **101** and the second guide slot **201**.

(73) A first actuator **400** may be configured with two driving motors but the number of driving motors is not limited to two and the first actuator **400** may be configured with one driving motor and coupled to a rear end of the first link part **100** or the second link part **200**.

(74) The second actuator **500** may be configured with one driving motor and move the hinge **300**. The second actuator **500** may be disposed on a support **600** to move the hinge **300** forward or backward. However, the arrangement of the second actuator **500** is not limited thereto and may be supported by the first link part **100** or the second link part **200**, in a similar manner as shown in FIG. 1.

(75) Even when the first link part **100** and the second link part **200a** are each configured as a single link, the hinge **300** may be moved to a preset position to correspond to a shape of an object.

Specifically, when a large grip force and a narrow gap between fingers is required to grip an object such as a dish, the hinge **300** may be moved forward to be close to the object by the second actuator **500**. When a small grip force and a wide gap between the fingers are required to grip an object such as a cup, the hinge **300** may be moved backward to be away from the object by the second actuator **500**.

(76) Accordingly, even when the first actuator **400** is operated with the same driving force, the grip device **1a** may efficiently grip an object with a force transmission ratio suitable for a shape of the object.

(77) FIG. **6** is a block diagram of a robot device according to an embodiment of the disclosure.

(78) Referring to FIG. **6**, a robot device **1000** according to an embodiment of the disclosure may include the grip device **1**, a processor **700**, an image sensor **810**, a pressure sensor **820**, a distance sensor **830**, and a memory **900**.

(79) The grip device **1** may include a first finger **10**, a second finger **20**, and link parts (e.g., a first link part **100** and a second link part **200**). The link parts may be disposed to cross each other with respect to a hinge **300** and may support the first finger **10** and the second finger **20**. The grip device **1** may adjust a support point of the link parts by changing a position of the hinge **300** on the link parts.

(80) Specifically, the link parts may include a first link part **100** and a second link part **200** disposed to cross each other with respect to the hinge **300**. The first link part **100** may be provided with a first guide slot (e.g., the first guide slot **101** and/or the first guide slot **121**) and may support the first finger **10**. The second linkage portion **200** may be provided with a second guide slot (e.g., the second guide slot **201** and/or the second guide slot **221**), disposed to cross the first link part **100**, and support the second finger **20**.

(81) The hinge **300** is movable within the first guide slot and the second guide slot and may connect the first link part **100** and the second link part **200** at an intersection of the first link part **100** and the second link part **200**.

(82) The grip device **1** may further include a second actuator **500** configured to move the hinge **300** within the first guide slot **121** and the second guide slot **221**. The actuator **500** may include the connection member **501**, the connection member **512**, and the connection member **522** connected to the hinge **300** and each having a variable length.

(83) The link parts (e.g., the first link part **100** and the second link part **200**) of the grip device **1** may be each embodied as a four-bar linkage as shown in FIG. **1** but are not limited thereto and may be each embodied as a single link as shown in FIG. **5**. Parts of components of the grip device **1** that are the same as those of the above-described examples may be briefly described or a detailed description thereof may be skipped here.

(84) The processor **700** may control overall operations of the robot device **1000**. To this end, the processor **700** may include a central processing unit (CPU) or an application processor (AP). The processor **700** may be implemented as at least one general-purpose processor, a digital signal processor, an application specific integrated circuit (ASIC), a system-on-chip (SoC), a microcomputer (MICOM), a driver IC, or the like.

(85) The image sensor **810** may detect at least one of a shape or position of an object. Specifically, the image sensor **810** may be configured to photograph objects around the robot device **1000** and may be a camera. For example, the image sensor **810** may be implemented as a three-dimensional (3D) camera or a depth camera.

(86) The image sensor **810** may be disposed on a body part or a head part of the robot device **1000** but is not necessarily limited thereto and may be disposed on various positions, such as the first link part **100**, the second link part **200**, the first finger **10**, and the second finger **20**, on the grip device **1**.

(87) The pressure sensor **820** may measure pressure applied to at least one of the first finger **10** and the second finger **20**. For example, the pressure sensor **820** may include a piezoelectric element or a

load cell, convert pressure applied to at least one of the first finger **10** and the second finger **20** into an electrical signal and transmit the electrical signal to the processor **700**.

(88) The pressure sensor **820** may be disposed on a surface of the first finger **10** and a surface of the second finger **20** that face each other. Two of the pressure sensor **820** may be provided and disposed on both the first finger **10** and the second finger **20** or one pressure sensor **820** may be provided and disposed on the first finger **10** or the second finger **20**.

(89) When the grip device **1** grips an object, the pressure sensor **820** may detect pressure applied to the first finger **10** and the second finger **20** by the object, convert the pressure into an electrical signal, and transmit the electrical signal to the processor **700**.

(90) The distance sensor **830** may detect a distance to an object. Specifically, the distance sensor **830** may be embodied as a time-of-flight (ToF) sensor that includes a light-emitting part and a light-receiving part and that detects a distance between the distance sensor **830** and an object on the basis of a time required for light emitted from the light-emitting part to be reflected from the object and received by the light-receiving part.

(91) For example, the distance sensor **830** may be implemented as an infrared sensor that detects a distance to an object on the basis of a time required for an infrared ray to be reflected from the object and received after the infrared ray is emitted. However, the type of the distance sensor **830** is not necessarily limited thereto and may be embodied as various types of sensors, such as an ultrasonic sensor or a light wave detection and ranging (LiDAR) sensor, which are capable of detecting a distance to an object. In addition, the ToF sensor is only an example, and a type of a sensor of embodiments of the disclosure is not limited as long as it is capable of detecting a distance to an object.

(92) A control process of the robot device **1000** using the distance sensor **830** will be described in detail with reference to FIGS. 7A to 7F below.

(93) The memory **900** may store an operating system (OS) for controlling overall operations of the components of the robot device **1000**, and instructions or data related to the components in the memory **900**.

(94) Accordingly, the processor **700** may control a number of hardware or software components of the robot device **1000** by using various instructions or data stored in the memory **900**, process a command or data received from at least one of the other components by loading the command or data into a volatile memory, and store various types of data in a nonvolatile memory.

(95) In particular, the memory **900** may store information about the shapes of various objects, a critical pressure value applied to the first finger **10** and the second finger **20** due to an object to be gripped, and the sum of distances detected by the distance sensor **830** at each point at which the grip device **1** is located.

(96) The processor **700** may receive and analyze information about the shape and position of an object detected by the image sensor **810** to identify the position, size, shape, and type of the object. Specifically, the processor **700** may identify the object through an object recognition algorithm to identify the size, shape, type, etc. of the object, and identify the position of the object on the basis of depth information of the object.

(97) Hereinafter, a control process of the robot device **1000** using the image sensor **810** according to an embodiment of the disclosure will be described in detail.

(98) The image sensor **810** may detect the position and shape of an object to be gripped and transmit information about the detected position and shape to the processor **700**. The processor **700** compares the information received from the image sensor **810** with the information about the shapes of various objects stored in the memory **900** to identify the type and position of the object.

(99) Thereafter, the processor **700** may control the grip device **1** to position the hinge **300** to correspond to the object detected by the image sensor **810**. Specifically, the processor **700** may determine lengths of the connection member **501**, the connection member **512**, and the connection member **522** according to at least one of the shape or position of the object detected by the image

sensor **810**, and control the actuator **500** such that the connection member **501**, the connection member **512**, and the connection member **522** have the determined length.

(100) For example, when the processor **700** identifies that an object to be gripped has a flat shape similar to a shape of a dish, the processor **700** may control the actuator **500** to increase the lengths of the connection member **501**, the connection member **512**, and the connection member **522**. Accordingly, with reference to FIG. **1**, the hinge **300** is moved close to the object and thus the grip device **1** may easily grip even a flat object with a high force transmission ratio while reducing the gap L2 between the first finger **10** and the second finger **20**.

(101) When the processor **700** identifies an object to be gripped as a tall object such as a cup, the processor **700** may control the actuator **500** to reduce the lengths of the connection member **501**, the connection member **512**, and the connection member **522**. Accordingly, with reference to FIG. **2**, the hinge **300** is moved away from the object and thus the grip device **1** may easily grip even a tall object with a low force transmission ratio while increasing the gap L2 between the first finger **10** and the second finger **20**.

(102) That is, the robot device **1000** according to an embodiment of the disclosure is capable of moving the hinge **300**, which functions as a support point, close to or away from an object through a control process of the image sensor **810** and the processor **700** even when the same driving source is used. Accordingly, the robot device **1000** is capable of effectively gripping objects of various shapes by changing the gap L2 between the first finger **10** and the second finger **20** and a force transmission ratio according to the shapes of the objects.

(103) When the processor **700** identifies that an object to be gripped is located above or below a center line C of the grip device **1**, the processor **700** may control the actuator **500** such that lengths of the connection member **512** of the third driving motor **510** and the connection member **522** of the fourth driving motor **520** are different from each other.

(104) Specifically, when the object is located above the center line C of the grip device **1**, the processor **700** may control the actuator **500** such that the connection member **512** of the third driving motor **510** is longer than the connection member **522** of the fourth driving motor **520** as shown in FIG. **3**.

(105) In contrast, when the object is located below the center line C of the grip device **1**, the processor **700** may control the actuator **500** such that the connection member **512** of the third driving motor **510** is shorter than the connection member **522** of the fourth driving motor **520** as shown in FIG. **4**.

(106) Accordingly, the grip device **1** is capable of easily performing a grip operation in a wide range without interfering with a support, such as a shelf, a floor, or a desk, with respect to a grip line G with a height different from that of the center line C.

(107) Hereinafter, a control process of the robot device **1000** using the pressure sensor **820** according to an embodiment of the disclosure will be described in detail.

(108) When an object to be gripped is positioned between the first finger **10** and the second finger **20**, the grip device **1** may perform a grip operation by the first actuator **400**. In this process, pressure may be applied to the first finger **10** and the second finger **20** by the object, and the pressure sensor **820** may measure a pressure value.

(109) The processor **700** may control the grip operation of the grip device **1** on the basis of the pressure value measured by the pressure sensor **820**. Specifically, the processor **700** may compare the pressure value measured by the pressure sensor **820** with a critical pressure value stored in the memory **900** and control an operation of the first actuator **400** to be stopped when the measured pressure value reaches the critical pressure value. According to embodiments, the processor **700** may compare the pressure value measured by the pressure sensor **820** with a plurality of pressure values stored in the memory **900**. For example, the plurality of pressure values may include a minimum pressure value and a maximum pressure value, and the processor **700** may control an operation of the first actuator **400** to reduce a gripping force when the measured pressure value is

above the maximum value and to increase a gripping force when the measured pressure value is below the minimum value.

(110) Accordingly, the robot device **1000** according to an embodiment of the disclosure may prevent an object from being broken due to excessively high pressure or from slipping from the grip device **1** due to excessively low pressure, and may stably grip the object with an appropriate grip force.

(111) A process in which an operation of the grip device **1** is controlled according to a result of detection of the distance sensor **830** will be described in detail with reference to FIGS. 7A to 7F below. FIGS. 7A to 7F are diagrams for describing a process in which a grip operation of the grip device **1** is controlled according to a measurement value of the distance sensor **830**.

(112) Referring to FIGS. 7A to 7F, the distance sensor **830** may further include a first distance sensor **831** on the first finger **10** and a second distance sensor **832** on the second finger **20**.

(113) When the first distance sensor **831** detects an object S during the movement of the robot device **1000**, the processor **700** may activate the second distance sensor **832**, receive information about distances to the object S detected by the first distance sensor **831** and the second distance sensor **832** at multiple points according to the movement of the robot device **1000**, and control the grip operation of the grip device **1** on the basis of the received information.

(114) Specifically, as shown in FIG. 7A, during the movement of the robot device **1000** toward the object S to be gripped, the first distance sensor **831** may detect the object S in an activated state and the second distance sensor **832** may be in a deactivated state until the object S is detected by the first distance sensor **831**.

(115) Accordingly, the robot device **1000** according to an embodiment of the disclosure may remove an optical interference effect that may occur when no object is located between the first finger **10** and the second finger **20**, and may minimize power consumption and prevent load on the processor **700** by activating only the first distance sensor **831**.

(116) Thereafter, as shown in FIG. 7B, when the first distance sensor **831** detects an object, the processor **160** may activate the second distance sensor **832**. Accordingly, the processor **700** may receive information about distances to the object S to be gripped from the first distance sensor **831** and the second distance sensor **832**.

(117) In addition, the processor **700** may determine whether the difference between the distance between the first distance sensor **831** and the object S and the distance between the second distance sensor **832** and the object S is greater than a threshold value. Specifically, when it is determined that the difference is greater than the threshold value, the processor **700** may control at least one of the first finger **10** and the second finger **20** such that the difference value is equal to or less than the threshold value.

(118) For example, when the threshold value is set to 0, the processor **700** may control at least one of the first finger **10** and the second finger **20** such that the distance between the first distance sensor **831** and the object S is equal to the distance between the second distance sensor **832** and the object S.

(119) Accordingly, when the grip device **1** grips the object S, the first finger **10** and the second finger **20** may simultaneously approach the object S at the same speed to stably grip the object S.

(120) Thereafter, as shown in FIGS. 7C, 7D, and 7E, the robot device **1000** may continuously move in a direction toward the object S. The processor **700** may sequentially receive information regarding distances to the object S from the first distance sensor **831** and the second distance sensor **832** during the movement of the robot device **1000**.

(121) In this case, the processor **700** may determine a point at which the distance to the object S is at a minimum on the basis of the information about the distances received from the first distance sensor **831** and the second distance sensor **832**. When distance values received from the first distance sensor **831** and the second distance sensor **832** at multiple points decrease sequentially, the processor **700** may continuously move the robot device **1000** in the direction toward the object S.

(122) When distances values received from the first distance sensor **831** and the second distance sensor **832** decrease sequentially but increase at a point, the processor **700** may determine a point immediately before the point as a point at which a distance to the object S is at a minimum, and control the robot device **1000** to return to the point at which the distance to the object S is at a minimum.

(123) As shown in FIG. 7F, when the grip device **1** is located at a point at which the distances between the first distance sensor **831** and the object S and the second distance sensor **832** and the object S are at a minimum, the processor **700** may control the first actuator **400** to cause the grip device **1** to grip the object S. Specifically, the grip device **1** may grip the object S as a gap L1 between two driving motors of the first actuator **400** and a gap L2 between the first finger **10** and the second finger **20** are controlled to be small.

(124) In this way, the robot device **1000** of the disclosure may grip a central part or a thickest part of the object S by gripping the object S at a point at which the distance between the first distance sensor **831** and the object S is at a minimum or at a point at which the distance between the second distance sensor **832** and the object S is at a minimum. Accordingly, the robot device **1000** of the disclosure is capable of stably gripping the object S and may prevent the object S from falling or being broken caused when edges of the object S are gripped.

(125) Although example embodiments of the disclosure have been illustrated and described herein, it should be understood that the disclosure is not limited thereto, and various modifications may be made by those of ordinary skill in the art without departing from scope of the present disclosure.

Claims

1. A grip device comprising: a first finger; a second finger facing the first finger; a first link part comprising a first guide slot and configured to support the first finger; a second link part comprising a second guide slot, the second link part crossing the first link part and configured to support the second finger; a hinge configured to be moved within the first guide slot and the second guide slot and connect the first link part and the second link part at an intersection of the first link part and the second link part; a first actuator configured to move at least one of the first link part and the second link part to adjust a gap between the first finger and the second finger; and a second actuator configured to move the hinge within the first guide slot and the second guide slot.
2. The grip device as claimed in claim 1, wherein the first link part further comprises: a first linkage connected to a first point on the first finger; a second linkage connected to a second point on the first finger; and a third linkage connected to the second linkage and provided in parallel with the first finger, the second link part comprises: a fourth linkage connected to a first point on the second finger; a fifth linkage connected to a second point on the second finger and provided in parallel with the fourth linkage; and a sixth linkage connected to the fifth linkage and provided in parallel with the second finger, and the first guide slot is formed in the second linkage, and the second guide slot is formed in the fifth linkage.
3. The grip device as claimed in claim 2, wherein a length of the third linkage is equal to a distance between the first point and the second point on the first finger, and a length of the sixth linkage is equal to a distance between the first point and the second point on the second finger.
4. The grip device as claimed in claim 3, wherein the first actuator is configured to adjust a gap between the third linkage and the sixth linkage.
5. The grip device as claimed in claim 4, wherein the first actuator comprises: a first driving motor configured to move the third linkage in a first direction; and a second driving motor configured to move the sixth linkage in a second direction opposite to the first direction.
6. The grip device as claimed in claim 2, wherein the second actuator comprises: a first driving motor provided on the second linkage and configured to move the hinge along the first guide slot; and a second driving motor provided on the fifth linkage and configured to move the hinge along

the second guide slot.

7. The grip device as claimed in claim 6, wherein each of the first driving motor and the second driving motor comprises a Scotch yoke type driving motor, a rack and pinion type driving motor, or a hydraulic type driving motor.

8. The grip device as claimed in claim 2, wherein the second linkage and the fifth linkage each have a convex shape with respect to a space between the first finger and the second finger.

9. The grip device as claimed in claim 1, wherein the second actuator comprises a driving motor configured to move the hinge within the first guide slot and the second guide slot.
